
Lecture Notes

on Visual Display of Data

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1 Frequency Distribution

1.0.1 Definition

A frequency distribution is a way of organizing data so that we can see how often each value (or group of values) occurs.

Instead of looking at raw numbers in a long list, the data are grouped into categories or intervals, and the number of times each category appears (its frequency) is recorded.

2 Types of Frequency Distribution

2.1 Tabular Form

This shows the data in a table with categories (or intervals) and their corresponding frequencies.

👉 **Examples:** test scores of 20 students

Score Range	Frequency
0–10	2
11–20	4
21–30	6
31–40	5
41–50	3

Table 1: Example of a frequency distribution in tabular form

This table tells us, for example, that 6 students scored between 21 and 30.

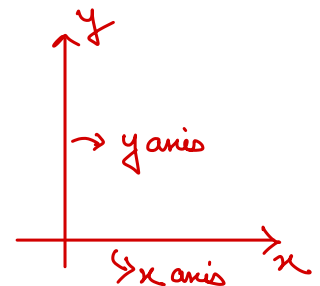
2.2 Graphical Form

Here the frequencies are shown with visuals. Common types include:

2.2.1 Bar Chart

A bar chart is a graphical way of showing a frequency distribution. It uses rectangular bars of equal width, where the height (or length) of each bar represents the frequency of a category or group.

- Bar charts are best for categorical data (like favorite colors, fruits, or movie genres).



- The categories are placed along the x-axis, and the frequencies are shown on the y-axis.
- Bars are separated by gaps (unlike histograms, where bars touch).

☛ **Examples:** Suppose we surveyed 10 students about their favorite fruit. The list is:

Apple, Orange, Banana, Apple, Mango, Apple, Banana, Mango, Apple, Orange

From this list, first we create a frequency table:

Fruit	Frequency
Apple	4
Orange	2
Banana	2
Mango	2

Table 2: Frequency distribution of favorite fruits

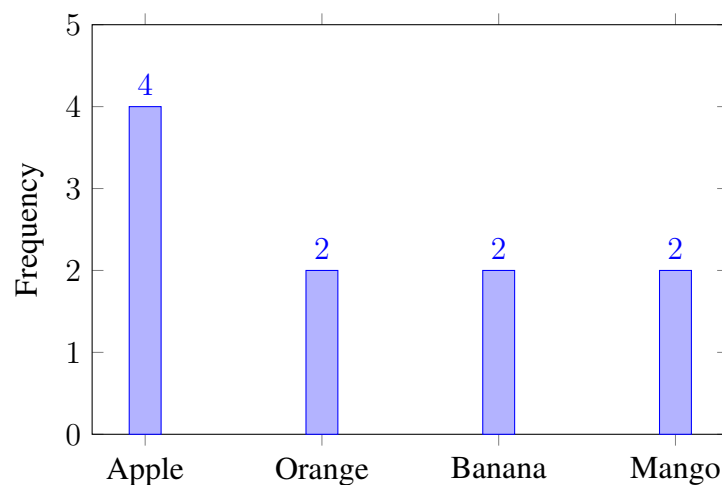


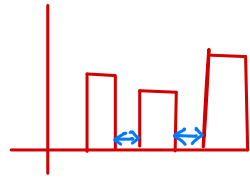
Figure 1: Bar chart of favorite fruits

2.2.2 Histogram

A histogram is a graphical way of showing a frequency distribution for numerical data. Unlike bar charts (which are for categories like fruits), histograms group numbers into intervals (called bins or classes), and the bins touch each other to show the continuous nature of the data.

- The x-axis shows the intervals (e.g., score ranges).
- The y-axis shows the frequency (how many data points fall in each interval).
- Bins are adjacent (no gaps) because the data are numerical and continuous.

How to construct bin size or range for a Histogram:



- (1) Decide on the number of classes (bins). If the dataset is large, choose around 10 to 20 classes; if the dataset is small, choose around 4 to 6 classes.

Thumb rule: Number of bins = $\frac{\text{number of observations}}{\text{desired class size (at least 4)}}$

- (2) Compute the width of each class:

$$\text{Class width} = \frac{\text{Range of data}}{\text{Number of classes from Step 1}}$$

Always round this result up to a convenient number.

- (3) Select the smallest data value (or a convenient smaller value) as the lower limit of the first class. Add multiples of the class width from Step 2 to generate the lower limits of the remaining classes.
- (4) Find the upper class limits by adding the class width to each lower limit, then subtract the smallest **significant unit** in the data (for whole numbers, subtract 1). This avoids overlap of class intervals.
- (5) To define class *boundaries*, take the **midpoint between the upper limit of one class and the lower limit of the next class**. This ensures the intervals touch and all data values are included without ambiguity.

☛ **Examples:** Suppose 20 students had the following test scores:

12, 25, 33, 41, 27, 38, 45, 21, 29, 19, 10, 30, 22, 36, 40, 14, 18, 32, 47, 24

Step (1): Number of classes.

There are $n = 20$ observations. Since the data are small, we choose 5 classes.

Step (2): Class width.

$$\text{Range} = 47 - 10 = 37.$$

$$\text{Class width} = \frac{37}{5} = 7.4 \Rightarrow \text{round up to } 8.$$

Step (3): Lower limits.

Start at the minimum value 10. The successive lower limits are:

10, 18, 26, 34, 42.

Step (4): Upper class limits.

Rule: upper limit = lower limit + width - smallest unit (= 1 here).

e.g., $10 + (8 - 1) = 10 + 7 = 17$

10 → 17,
18 → 25,
26 → 33,
34 → 41,
42 → 49.

$10, 10+8, 10+8+8, 10+8+8+8, 10+8+8+8+8$
 $10, 18, 26, 34, 42$
 $10 + (2 \times 8) \quad 10 + (3 \times 8)$
 $10 + 8 + 8 + 8 + 8$
 $10 + (4 \times 8)$
 Total 5 as we considered 5 classes.
 So once we get the interval say for first class [10, 17]

Here $\frac{1}{2} \times 1 = 0.5$
 $\therefore$ Boundaries will be [10, 17]
 $-0.5 \quad +0.5$

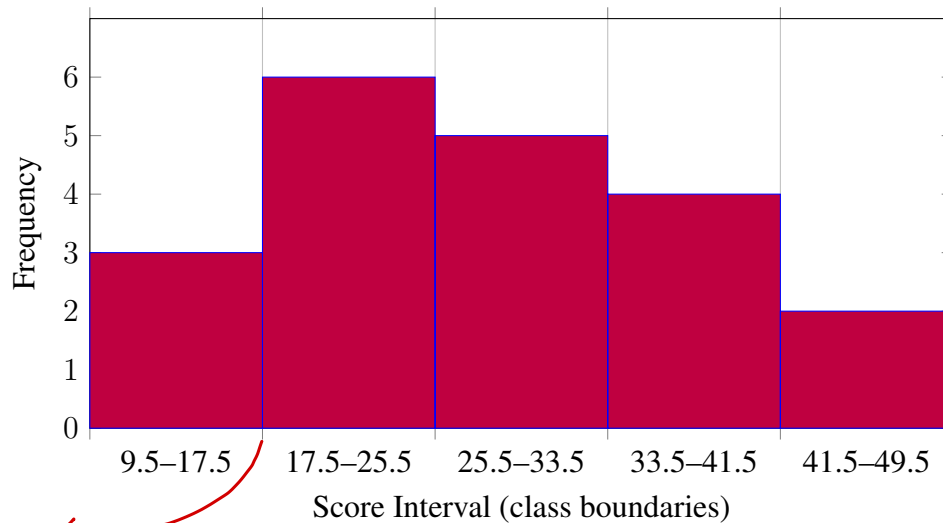
Step (5): Class boundaries.

Apply the ± 0.5 rule (halfway between limits):

$[10, 17]$
 $\rightarrow [10 - 0.5, 17 + 0.5]$
 $\rightarrow [9.5, 17.5], [17.5, 25.5], [25.5, 33.5], [33.5, 41.5], [41.5, 49.5]$.

Frequency Table:

Class Interval (limits)	Class Boundaries	Frequency
10–17	9.5–17.5	3
18–25	17.5–25.5	6
26–33	25.5–33.5	5
34–41	33.5–41.5	4
42–49	41.5–49.5	2



Now no issue to share

the common boundary (17.5 here) as no data pt will be 17.5 since 1 is the significant unit.

2.2.3 Scatter Plots

A scatter plot is a graph that shows the relationship between two variables.

- Each data point is shown as a dot on the coordinate plane:
 - 1 The x-axis represents one variable.
 - 2 The y-axis represents the other variable.
- By looking at the pattern of points, we can see if the variables are related.

Uses of scatter plots:

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- To check if two variables have a positive relationship (as one increases, the other increases).
 - To check for a negative relationship (as one increases, the other decreases).
 - To detect if there is no clear relationship (points scattered randomly).
 - To identify possible **outliers** (points far away from the rest).

👉 **Examples:**

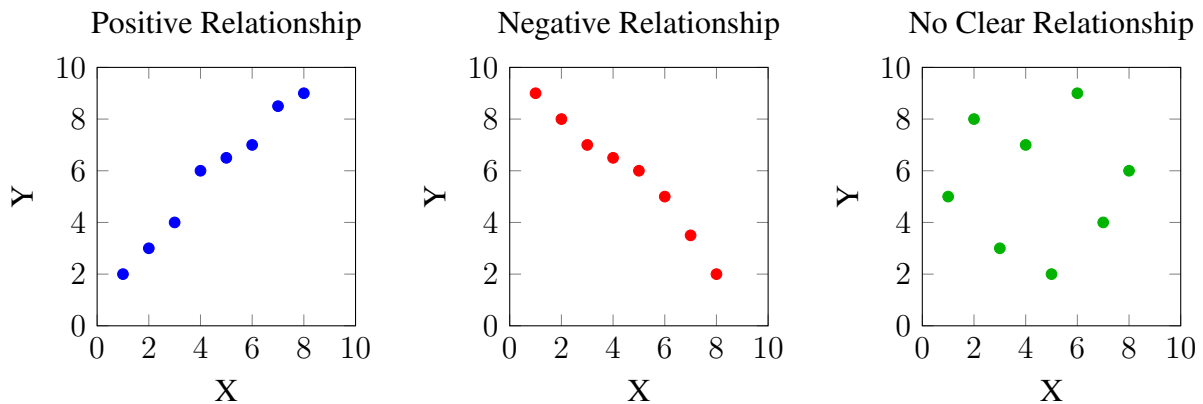


Figure 3: Scatter plots showing positive (blue), negative (red), and no relationship (green) between two variables